Can we predict which open Erdős problems AI will help resolve?

Preregistration SHA-256:
36aa4ca93b1e8730dae0573b1c2523163ba0a54ab213be19b5ef8c1b5b76306a
  PREREGISTRATION.md 2026-10-22 02:44:51 EDT

Backtest on 7 monthly origins 2026-01-01 to 2026-07-01, 3-month windows.
Report built 2026-10-22 (US Eastern).

H1: NO

Obscurity and attention features alone rank solved problems above unsolved ones with AUC 0.571 (95% CI 0.511 to 0.632); the preregistered bar was above 0.60. [medium]

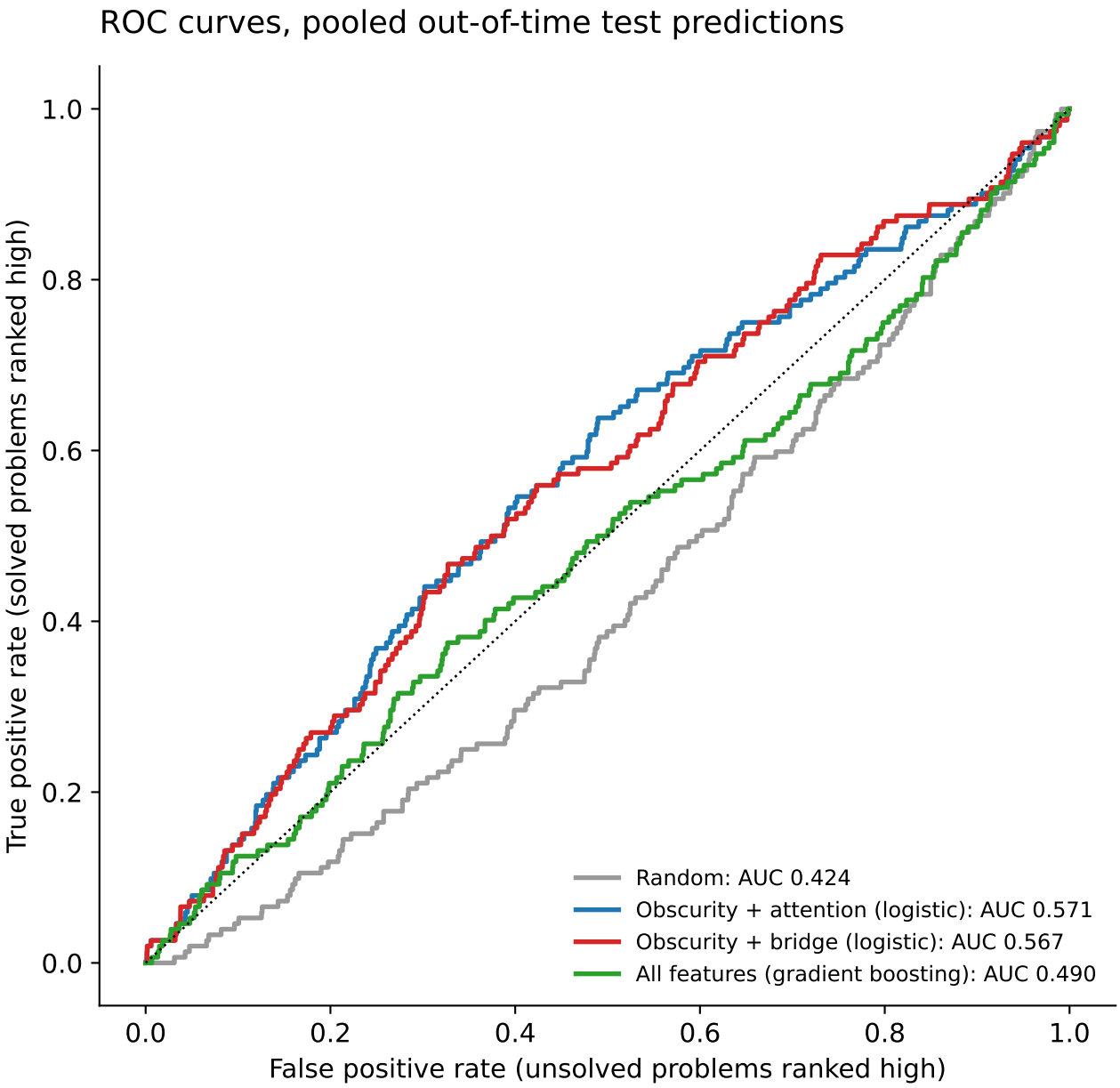
H2: NO

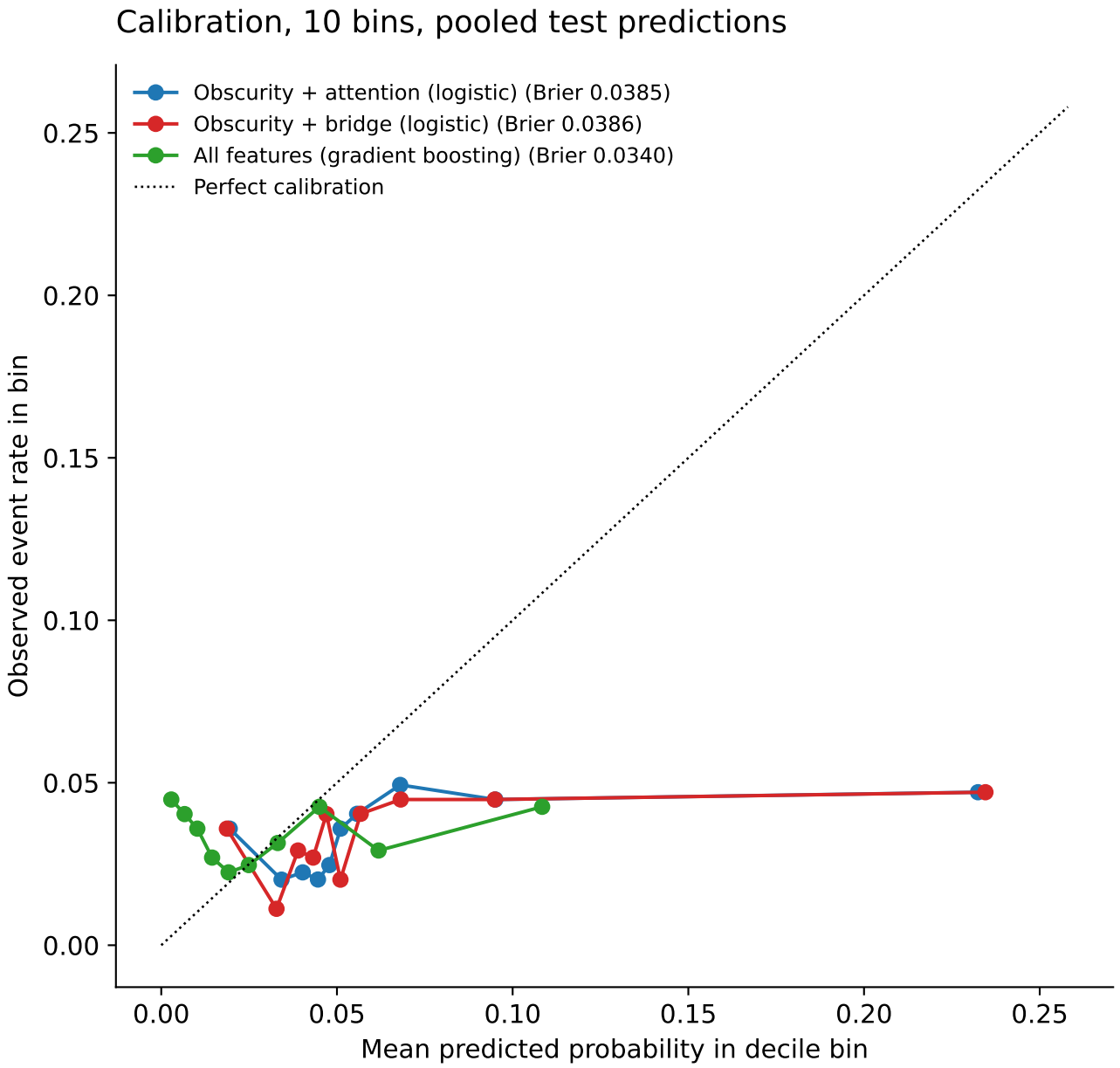
Adding bridge distance moves AUC from 0.571 to 0.567 (change -0.004).
DeLong p = 0.540; permutation p = 0.615;
problem-clustered bootstrap 95% CI of the change: -0.024 to +0.015. [medium]

AUC = chance the model ranks a random solved problem above a random unsolved one (0.5 = coin flip).

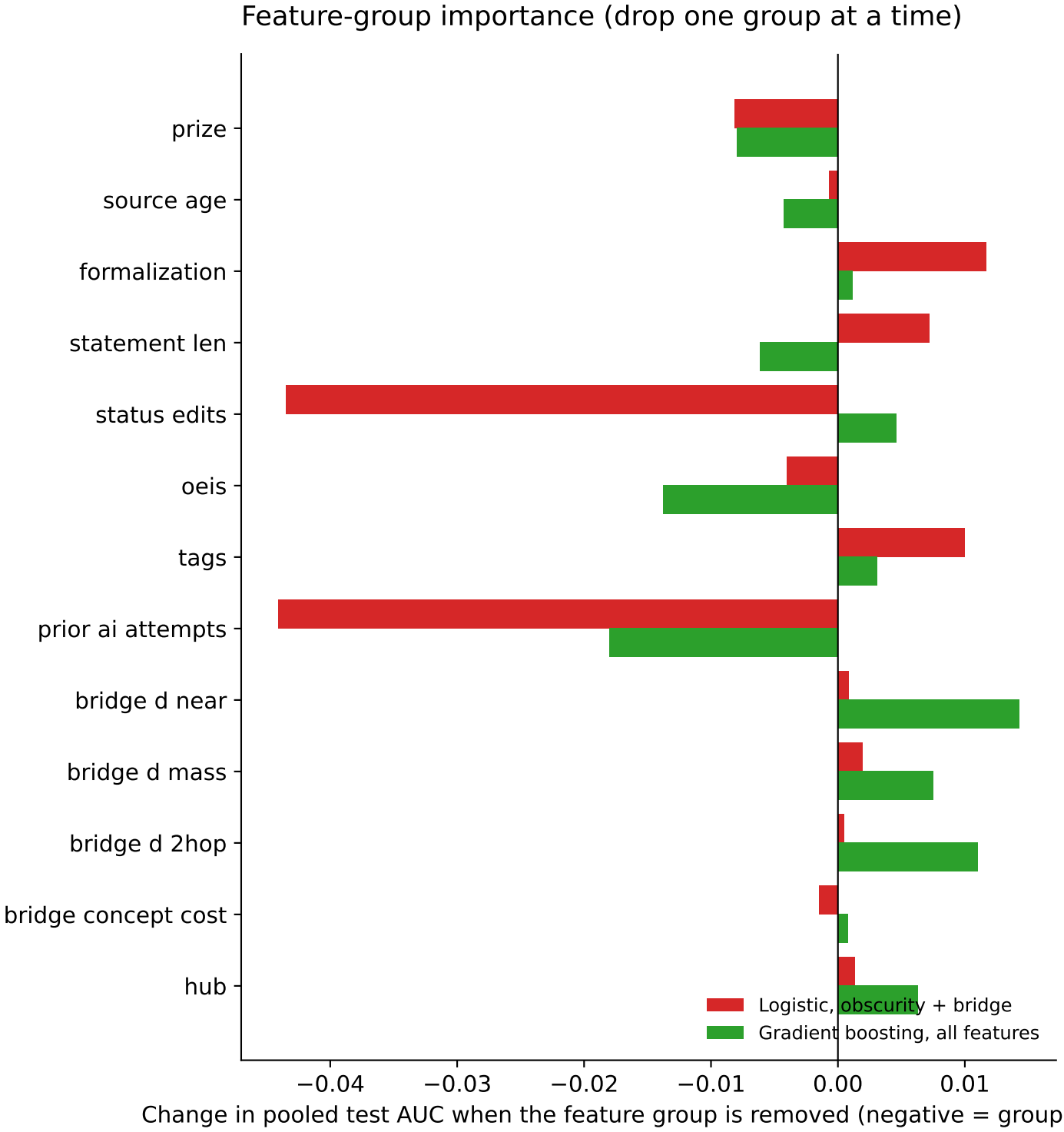
Pooled test rows: 4458, events: 152 (base rate 0.034).

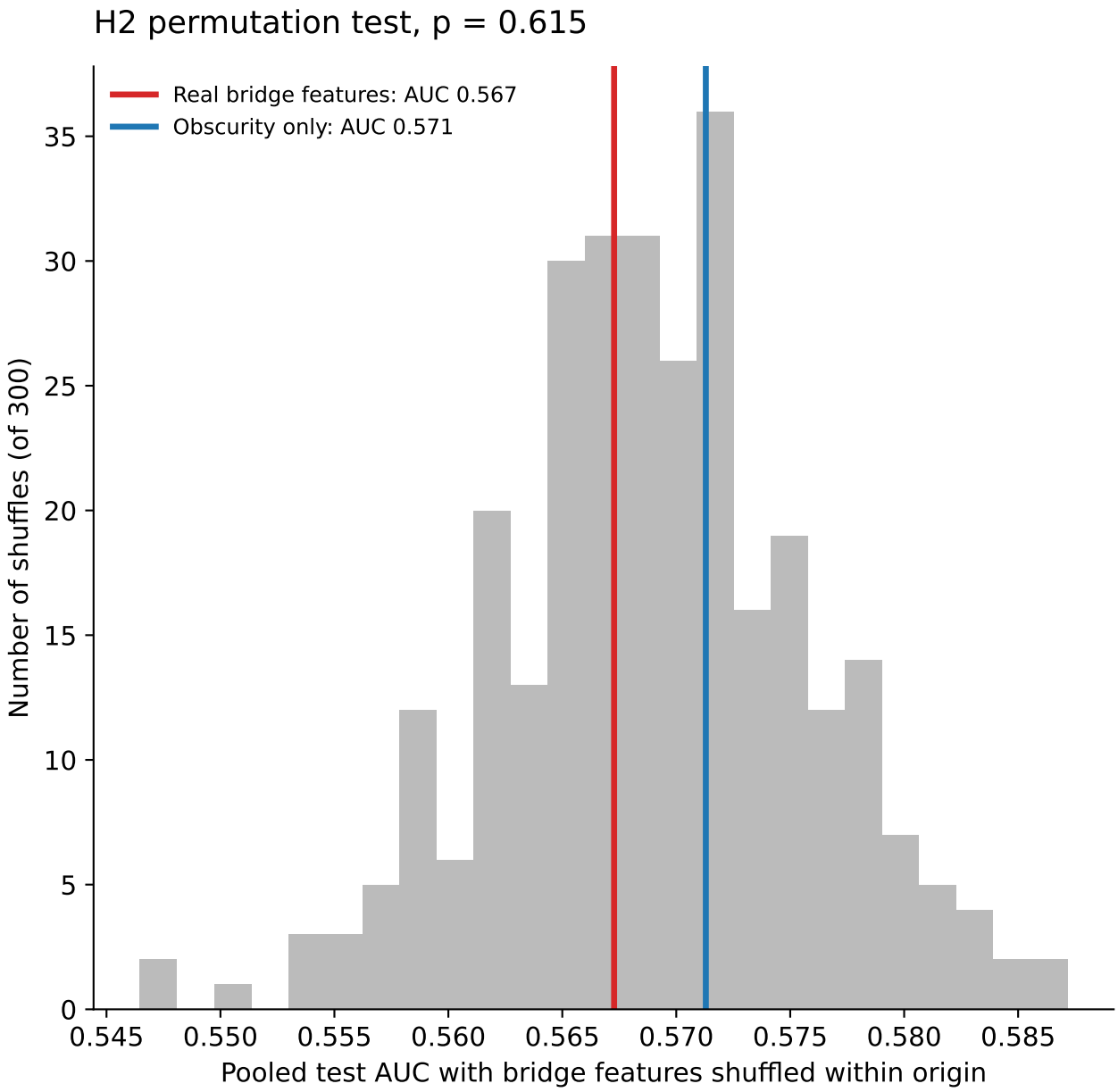
A null H2 is a finding: within this data, how close a problem sits to existing arXiv mathematics does not add detectable signal beyond obscurity and attention.


ROC curves, pooled out-of-time test predictions
True positive rate (solved problems ranked high)
False positive rate (unsolved problems ranked high)
0.0
0.2
0.4
0.6
0.8
1.0
Random: AUC 0.424
Obscurity + attention (logistic): AUC 0.571
Obscurity + bridge (logistic): AUC 0.567
All features (gradient boosting): AUC 0.490



Brier of always predicting the base rate (0.034): 0.0329. Precision@20 per origin: Random 0.00, Obscurity + attention 0.09, Obscurity + bridge 0.09, All features 0.04


Feature-group importance (drop one group at a time)
prize
source age
formalization
statement len
status edits
oeis
tags
prior ai attempts
bridge d near
bridge d mass
bridge d 2hop
bridge concept cost
hub
Logistic, obscurity + bridge
Gradient boosting, all features
−0.04
−0.03
−0.02
−0.01
0.00
0.01
Change in pooled test AUC when the feature group is removed (negative = group helped


H2 permutation test, p = 0.615
Real bridge features: AUC 0.567
Obscurity only: AUC 0.571
Number of shuffles (of 300)
0
5
10
15
20
25
30
35
0.545
0.550
0.555
0.560
0.565
0.570
0.575
0.580
0.585
Pooled test AUC with bridge features shuffled within origin

Forecast: top 20 open problems, 2026-10-22

Model: All features (gradient boosting). Expected AI-assisted resolutions among 593 open problems in 3 months: 7.8. 12-month column assun
constant hazard [low].

Rank	Problem	P(3 mo)	P(12 mo) [low]	Statement (start)
1	11	0.083	0.293	Is every large odd integer n the sum of a squarefree number and [...]
2	873	0.081	0.286	Let A=\{a_1<a_2<\cdots\}\subseteq \mathbb{N} and let F(A,X,k) [...]
3	1183	0.079	0.280	Let f(n) be maximal such that in any 2-colouring of the subsets [...]
4	75	0.077	0.274	Is there a graph of chromatic number \aleph_1 with \aleph_1 [...]
5	408	0.073	0.261	Let \phi(n) be the Euler totient function and \phi_k(n) be the [...]
6	1194	0.073	0.261	Let A\subset\mathbb{N} be such that every integer n\geq 1 can be [...]
7	749	0.070	0.251	Let \epsilon>0. Does there exist A\subseteq \mathbb{N} such that [...]
8	52	0.065	0.236	Let A be a finite set of integers. Is it true that for every [...]
9	595	0.064	0.231	Is there an infinite graph G which contains no K_4 and is not [...]
10	1188	0.060	0.219	Call a set of distinct integers 1<n_1<\cdots<n_k with associated [...]
11	1143	0.058	0.213	Let p_1<\cdots<p_u be primes and let k\geq 1. Let [...]
12	12	0.057	0.210	Let A be an infinite set such that there are no distinct [...]
13	1201	0.056	0.207	Is it true that for every \epsilon,\eta>0 there exists a k such [...]
14	396	0.056	0.207	Is it true that for every k there exists n such [...]
15	1209	0.054	0.200	Let A=\{a_1<a_2<\cdots\} be a sequence of integers which tends [...]
16	1151	0.052	0.193	Given a_1,\ldots,a_n\in [-1,1] let\[\mathcal{L}^nf(x) = [...]
17	550	0.046	0.172	Let m_1\leq\cdots\leq m_k and n be sufficiently large. If T is a [...]
18	1218	0.046	0.171	Assume the Generalised Continuum Hypothesis and let [...]
19	1207	0.046	0.170	Let P_d(n) be such that in any set of n points in \mathbb{R}^d [...]
20	1206	0.045	0.169	Does \{1,2^3,\ldots,N^3\} contain a Sidon set of size \gg N? Is [...]

Secondary analyses: literature identification vs new solution

```
lit_id   lr_obs           AUC 0.743  PR-AUC 0.010  base 0.003  events 14

lit_id   lr_obs_bridge    AUC 0.736  PR-AUC 0.009  base 0.003  events 14

lit_id   hgb_all          AUC 0.475  PR-AUC 0.006  base 0.003  events 14

new_sol  lr_obs           AUC 0.547  PR-AUC 0.051  base 0.031  events 138

new_sol  lr_obs_bridge    AUC 0.539  PR-AUC 0.048  base 0.031  events 138

new_sol  hgb_all          AUC 0.479  PR-AUC 0.031  base 0.031  events 138
```

Side test D (Haiku technique rating), logistic obscurity + D: AUC 0.569 vs obscurity alone 0.571. Contamination probe: events=68 acc_events=0.059 comp=0.073 fpr_unresolved=0.088. Not part of any headline model.

Per-type counts are small; AUC differences under about 0.05 are within noise [low].

Post hoc check (exploratory, not part of H1 or H2)

Same features, all 11 origins pooled (7012 rows, 229 events), problem-grouped 5-fold cross-validation. This uses more training data but is not out-of-time, so it can leak time trends. Run after the backtest; it changes no conclusion.

AUC: obscurity logistic 0.556; obscurity + bridge logistic 0.548; gradient boosting with bridge 0.619.

Bridge effect: -0.007 AUC (DeLong p 0.13; clustered 95% CI -0.024 to +0.009). Bridge features still add nothing detectable [low].

Limitations

Few events, short history. Status history exists only from 2025-08-31, so training data is 4 origins (Sep to Dec 2025) before the first test origin; the January model trains on 2 origins. Test events are 152 rows from about 68 problems because windows overlap [high].

Label noise. "AI involvement" comes from a volunteer wiki, with a 120-day lookback and 30-day lookahead rule I chose. Resolutions are maintainers' status edits, which lag the mathematics. Wiki selection bias: attempts are listed more often when they succeed [medium].

Statement text and tags on the site are current versions, not versions as of each origin [medium]. Reference lists were excluded for the same reason. OpenAlex citation counts were not built.

Bridge distance uses MiniLM-L6 embeddings of arXiv abstracts. A weak embedding of LaTeX statements could hide real signal, so a null H2 limits this implementation, not the idea [medium].

Rows repeat the same problem across up to 3 origins. DeLong assumes independent rows, so its p is optimistic; use the clustered bootstrap and permutation figures [medium].

The 12-month probability assumes a constant monthly hazard, but AI capability and attention are rising [low]. Wiki ends 2026-06-30, so July 2026 labels depend only on status changes after that date [medium].

Method and data

Data: teorth/erdosproblems problems.yaml git history (monthly snapshots), AI-contributions wiki tables (dates, outcomes), erdosproblems.com statements (1,221 of 1,221), google-deepmind/formal-conjectures git dates, arXiv metadata snapshot (librarian-bots/arxiv-metadata-snapshot, 739,239 math and relevant cs papers).

Labels: event = open at origin, resolved within 3 months, with a green or yellow AI wiki row in sections 1a to 1d or 2a near the resolution date. Hand audit of 25 non-trivial labels: 25 of 25 consistent with the wiki.

Models, features, origins and metrics were fixed in PREREGISTRATION.md (hash on page 1) before any test metric was computed.

Code and forecast: https://github.com/Questboy23/ai-math-forecast. forecast.csv SHA-256: 2c9a9d86ec22b205717723ce04ad9d8d6e013372f0f79b897e6813b3f1245981.